# Tecnológico Nacional de México

## Instituto Tecnológico Superior de Felipe Carrillo Puerto

## Ingeniería en Sistemas Computacionales

**Asignatura**
Programación Web del lado del Servidor

**Tema**
2 Desarrollo de API´s

**EVIDENCIA DE APRENDIZAJE**
AA 2.1 API Reporte Práctico

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**ISC-8B**

Felipe Carrillo Puerto a 23 de marzo de 2026

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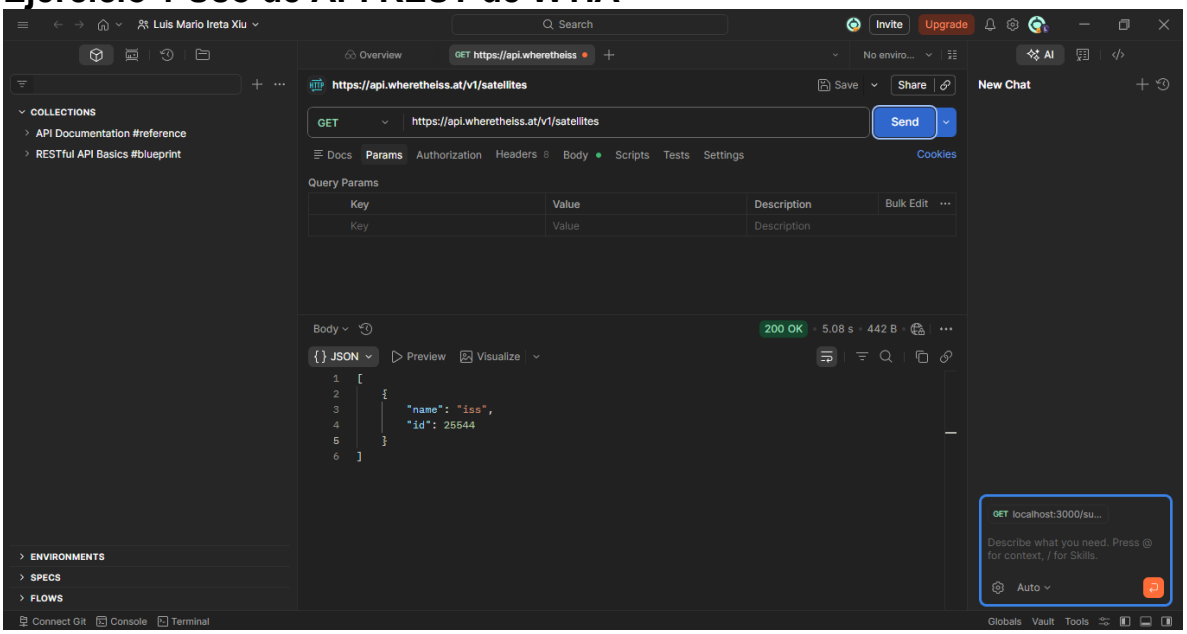
# Introducción APIs

## Ejercicio 1 Uso de API REST de WTIA



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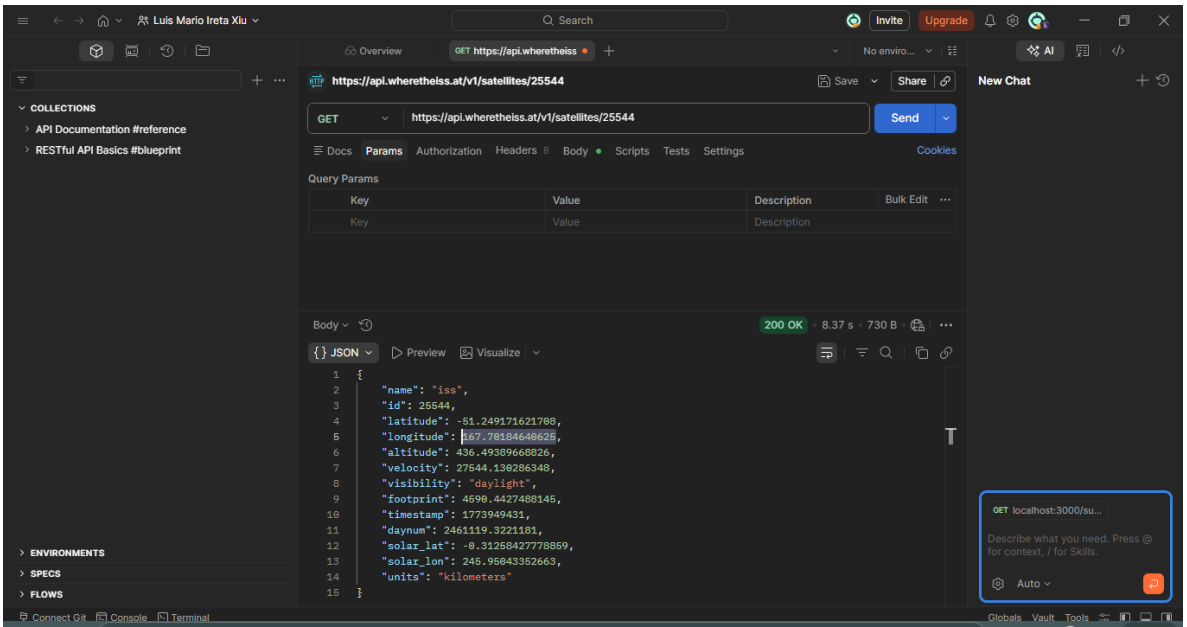


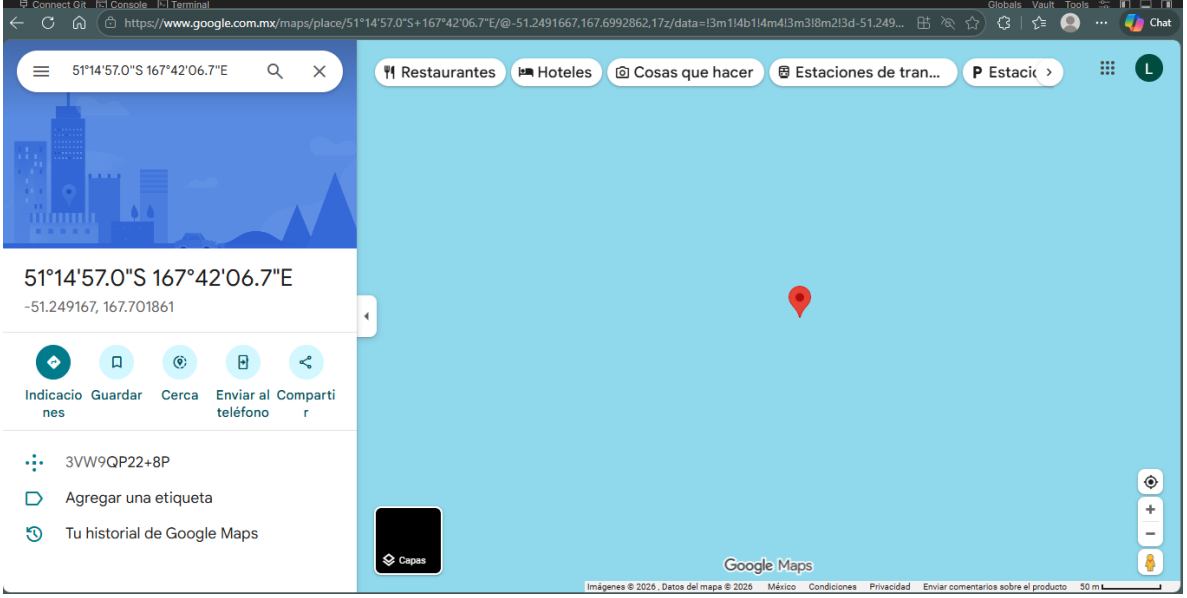




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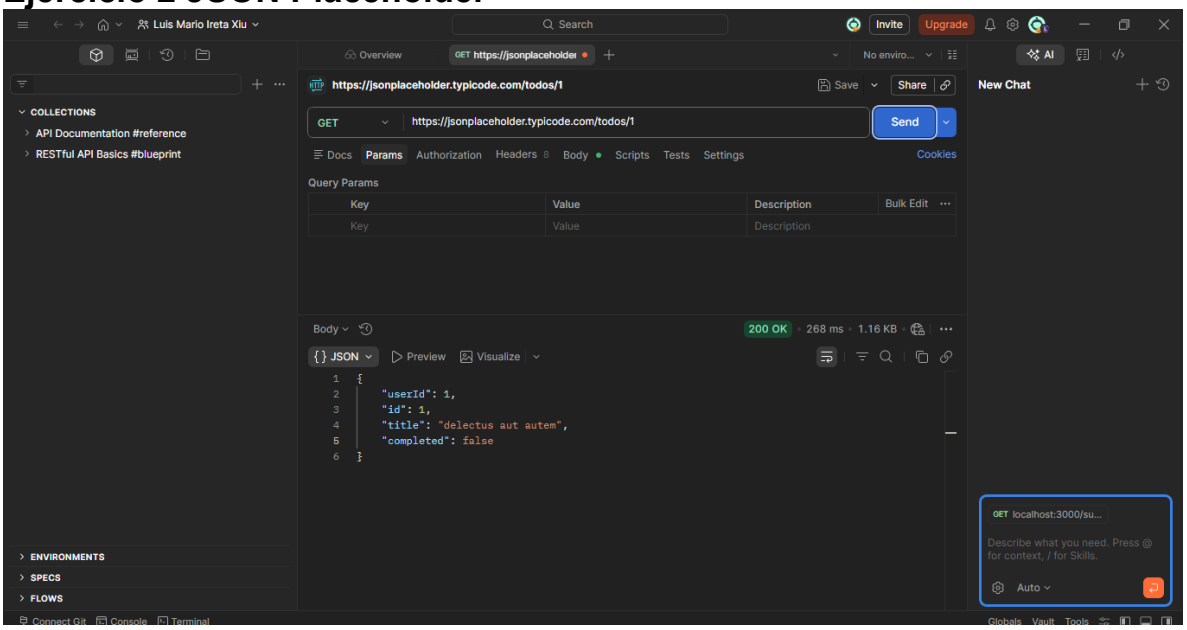
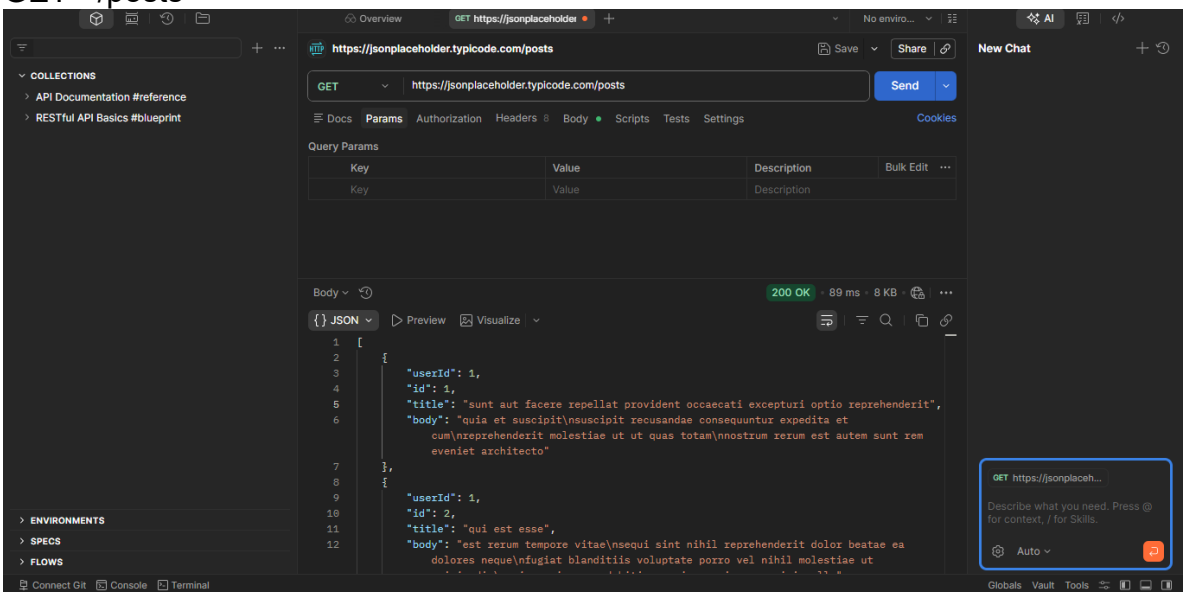
## Ejercicio 2 JSON Placeholder




## Pruebas de métodos con Postman

GET /posts



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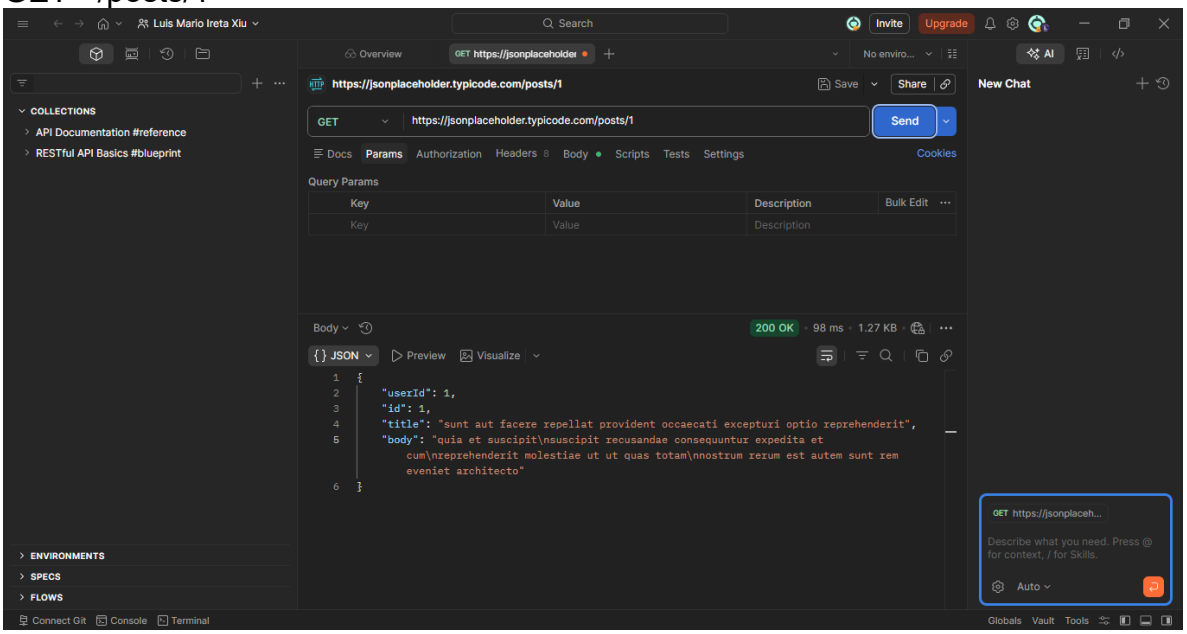
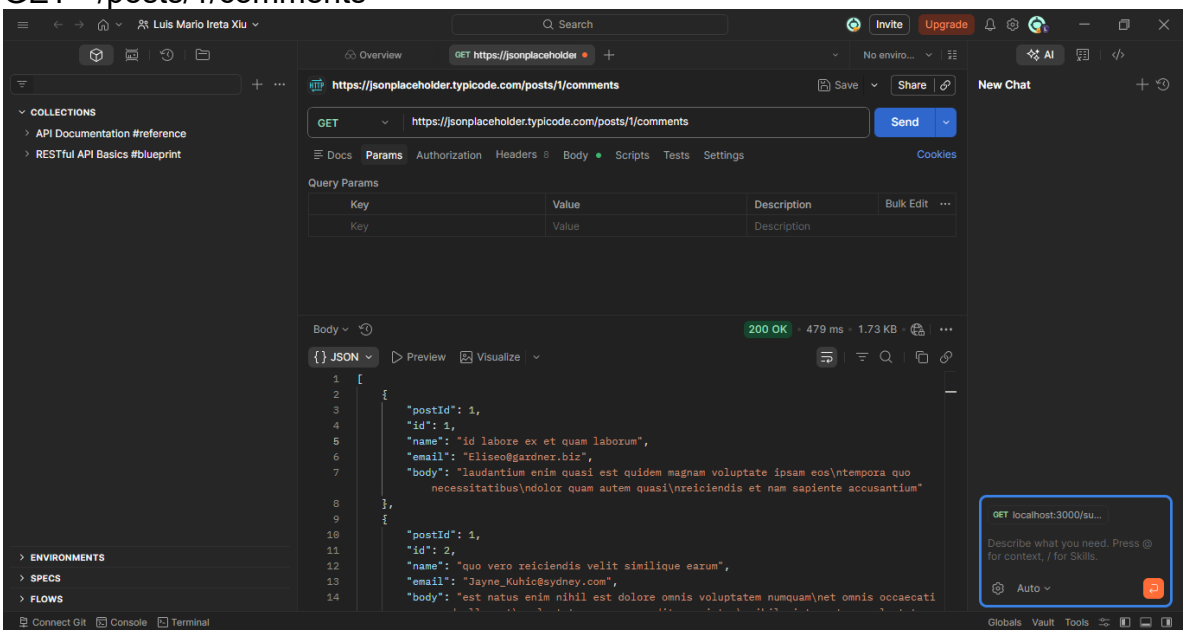
GET /posts/1




GET /posts/1/comments



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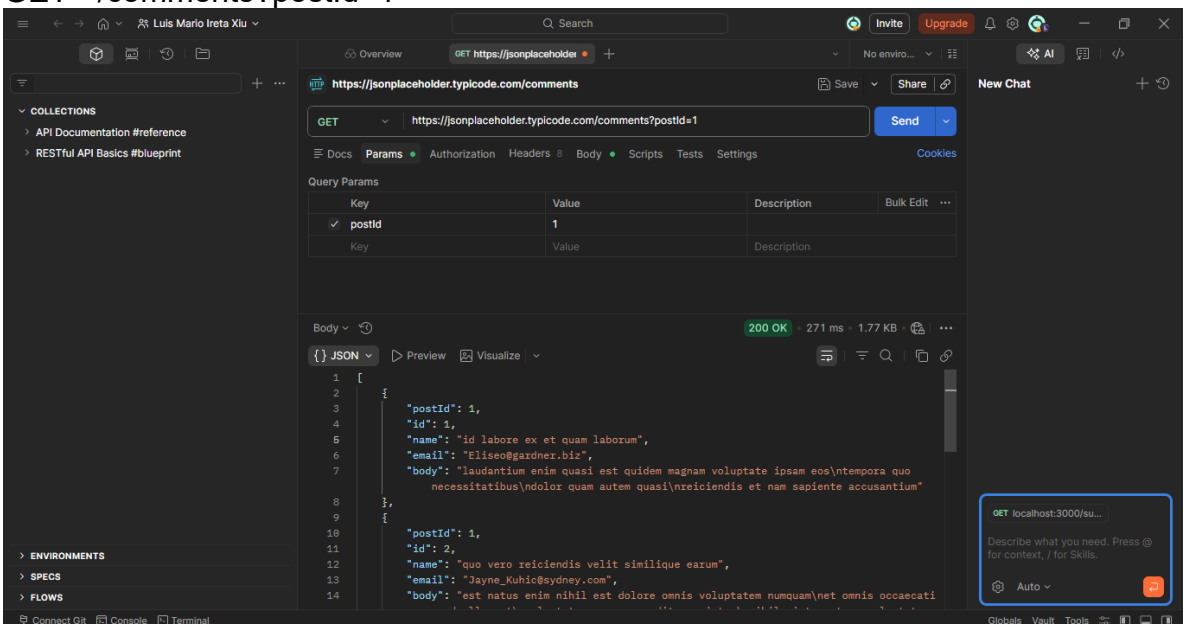
GET /comments?postId=1



## Ejercicio 3 JSON de la información ipquery


```
{
    "ip": "187.155.247.151",
    "isp": {
        "asn": "AS8151",
        "org": "Uninet S.A. de C.V.",
        "isp": "Uninet S.A. de C.V."
    },
    "location": {
        "country": "Mexico",
        "country_code": "MX",
        "city": "Cancún",
        "state": "Quintana Roo",
        "zipcode": "",
        "latitude": 21.181205791757332,
        "longitude": -86.86995173872624,
        "timezone": "America/Cancun",
        "localtime": "2026-03-19T20:13:58"
    },
    "risk": {
        "is_mobile": false,
        "is_vpn": false,
        "is_tor": false,
        "is_proxy": false,
        "is_datacenter": false,
        "risk_score": 0
    }
}
```

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## Formato de peticiones

### Endpoints.

URLBase/**endpoint**
https://randomuser.me/api/

https://randomuser.me/api/ Chat

Impresión con sangría


```
{"results":[{"gender":"male","name":
{"title":"Mr","first":"Juan","last":"Martínez"},"location":{"street":
{"number":5466,"name":"Paseo de
Extremadura"},"city":"Valladolid","state":"País
Vasco","country":"Spain","postcode":36981,"coordinates":
{"latitude":"-37.3990","longitude":"-116.1983"},"timezone":
{"offset":"+3:00","description":"Baghdad, Riyadh, Moscow, St.
Petersburg"}},"email":"juan.martinez@example.com","login":
{"uuid":"a888eff4-d22c-4cc2-87c4-
97926fbe23e7","username":"smallfish651","password":"jazz","salt":"YikXVU
jb","md5":"91f7445f07261ce1ee82800da2e48d81","sha1":"9eb5fde3cfa6277f1f6
d35eb2b3d407f17fa262","sha256":"e919525a602adf519dc4f8568e37ef28c24be33
e3c9346bbb17f0ec852947a38"},"dob":{"date":"1969-09-
15T13:37:34.611Z","age":56},"registered":{"date":"2014-10-
27T09:50:51.963Z","age":11},"phone":"949-781-184","cell":"627-711-
050","id":{"name":"DNI","value":"28162486-W"},"picture":
{"large":"https://randomuser.me/api/portraits/men/12.jpg","medium":"http
s://randomuser.me/api/portraits/med/men/12.jpg","thumbnail":"https://ran
domuser.me/api/portraits/thumb/men/12.jpg"},"nat":"ES"}],"info":
{"seed":"6922e6ff29ba2284","results":1,"page":1,"version":"1.4"}}
```


https://randomuser.me/api/1.4/

https://randomuser.me/api/1.4/ Chat

Impresión con sangría


```
{"results":[{"gender":"male","name":
{"title":"Mr","first":"Jessie","last":"Jenkins"},"location":{"street":
{"number":4553,"name":"Camden Ave"},"city":"Port Macquarie","state":"New
South Wales","country":"Australia","postcode":9765,"coordinates":
{"latitude":"-87.2341","longitude":"-125.8897"},"timezone":
{"offset":"+5:00","description":"Ekaterinburg, Islamabad, Karachi,
Tashkent"}},"email":"jessie.jenkins@example.com","login":
{"uuid":"77a71a2d-cc3b-4d3c-bef0-
d87f4f95b4cb","username":"yellowbear862","password":"trigger","salt":"YX
nTYoW9","md5":"159403414d3a373a1d2a24caf9889563","sha1":"3a7d2a9284d495a
f816a63ed68decffa0cd55d70","sha256":"0d1fc647691067dae74258e6c400ea298a6
c865fb0bf8c57753a92d8477e8322"},"dob":{"date":"1976-04-
17T10:53:29.063Z","age":49},"registered":{"date":"2012-02-
11T08:45:08.632Z","age":14},"phone":"02-9537-2779","cell":"0492-670-
889","id":{"name":"TFN","value":"526268109"},"picture":
{"large":"https://randomuser.me/api/portraits/men/99.jpg","medium":"http
s://randomuser.me/api/portraits/med/men/99.jpg","thumbnail":"https://ran
domuser.me/api/portraits/thumb/men/99.jpg"},"nat":"AU"}],"info":
{"seed":"2c2432aae87562b5","results":1,"page":1,"version":"1.4"}}
```




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https://jsonplaceholder.typicode.com/comments


```
Impresión con sangría

[
  {
    "postId": 1,
    "id": 1,
    "name": "id labore ex et quam laborum",
    "email": "Eliseo@gardner.biz",
    "body": "laudantium enim quasi est quidem magnam voluptate ipsam
eos\ntempora quo necessitatibus\ndolor quam autem quasi\nreiciendis et
nam sapiente accusantium"
  },
  {
    "postId": 1,
    "id": 2,
    "name": "quo vero reiciendis velit similique earum",
    "email": "Jayne_Kuhic@sydney.com",
    "body": "est natus enim nihil est dolore omnis voluptatem
numquam\net omnis occaecati quod ullam at\nvoluptatem error expedita
pariatur\nnihil sint nostrum voluptatem reiciendis et"
  },
  {
    "postId": 1,
    "id": 3,
    "name": "odio adipisci rerum aut animi",
    "email": "Nikita@garfield.biz",
    "body": "quia molestiae reprehenderit quasi aspernatur\naut
expedita occaecati aliquam eveniet laudantium\nomnis quibusdam
delectus saepe quia accusamus maiores nam est\ncum et ducimus et vero
voluptates excepturi deleniti ratione"
  },
```


## Parámetros Query.

URLBase/endpoint/**query**
https://randomuser.me/api/?gender=female


```
Impresión con sangría

{"results":[{"gender":"female","name":
{"title":"Miss","first":"Chandni","last":"Molenaar"},"location":
{"street":{"number":6485,"name":"De
Wildtstraat"},"city":"Oosterwijtwerd","state":"Friesland","country":"Net
herlands","postcode":"9348 XY","coordinates":
{"latitude":"35.0581","longitude":"-127.6731"},"timezone":
{"offset":"-11:00","description":"Midway Island,
Samoa"}},"email":"chandni.molenaar@example.com","login":
{"uuid":"700a6a3a-60a5-49b4-819f-
044d8619a808","username":"organiczebra146","password":"aquarius","salt":
"ybPuJRsI","md5":"bc77463ed54a7065e1ea43432d241f34","sha1":"21a13eed4050
2dbd3569c5019a4f89047f2ae99b","sha256":"d5b1e838f73a98d9e6e0500fcf58efe9
217c69def61d4925a10cc4ddb20fba16"},"dob":{"date":"1960-09-
15T07:35:37.400Z","age":65},"registered":{"date":"2007-05-
23T16:12:45.668Z","age":18},"phone":"(0328) 554006","cell":"(06)
56218843","id":{"name":"BSN","value":"46217532"},"picture":
{"large":"https://randomuser.me/api/portraits/women/79.jpg","medium":"ht
tps://randomuser.me/api/portraits/med/women/79.jpg","thumbnail":"https:/
/randomuser.me/api/portraits/thumb/women/79.jpg"},"nat":"NL"}],"info":
{"seed":"7f0dabca353a176c","results":1,"page":1,"version":"1.4"}}
```




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https://randomuser.me/api/1.4/?gender=female&nat=MX

https://randomuser.me/api/1.4/?resu...

Impresión con sangría

{"results":[{"gender":"female","name":
{"title":"Ms","first":"Estela","last":"Monroy"},"location":{"street":
{"number":9908,"name":"Andador Indonesia"},"city":"San Antonio del
Mar","state":"Tamaulipas","country":"Mexico","postcode":55372,"coordin
ates":{"latitude":"-59.3384","longitude":"-52.6855"},"timezone":
{"offset":"+9:30","description":"Adelaide,
Darwin"}},"email":"estela.monroy@example.com","login":
{"uuid":"9eef0ad8-6304-4c57-96af-
74ac07369ea6","username":"redzebra504","password":"golfer1","salt":"xm
brVuCJ","md5":"cd7a866495f84af31f1107a06945f426","sha1":"719449c505659
f2c27ad474211a149dc8ee13678","sha256":"2703d975db99558e84cea5cc3151a57
5f0a3481ad67ec3077547932a124f4e9d"},"dob":{"date":"1962-05-
06T13:12:12.075Z","age":63},"registered":{"date":"2015-10-
22T18:19:00.772Z","age":10},"phone":"(604) 236 9632","cell":"(604) 073
0427","id":{"name":"NSS","value":"34 62 89 0731 2"},"picture":
{"large":"https://randomuser.me/api/portraits/women/73.jpg","medium":"
https://randomuser.me/api/portraits/med/women/73.jpg","thumbnail":"htt
ps://randomuser.me/api/portraits/thumb/women/73.jpg"},"nat":"MX"},
{"gender":"female","name":
{"title":"Miss","first":"Elvira","last":"Granados"},"location":
{"street":{"number":9432,"name":"Periférico Morelos"},"city":"Medellín
de
Bravo","state":"Guanajuato","country":"Mexico","postcode":13968,"coord
inates":{"latitude":"-45.9389","longitude":"-157.1253"},"timezone":
{"offset":"+5:30","description":"Bombay, Calcutta, Madras, New
Delhi"}},"email":"elvira.granados@example.com","login":
{"uuid":"60b6173b-57a4-4ef1-8524-
d560bfc750b3","username":"lazylion286","password":"saint","salt":"Roex
3ttm","md5":"612dfce124ad04732effd59676c47bd2","sha1":"4a17e356236a982
88c81ef4fb165c6f4de866c31","sha256":"42cd718d5a09bbce2013e85a948fe1517

https://randomuser.me/api/1.4/?results=10&gender=female&nat=MX

https://randomuser.me/api/1.4/...

Chat

Impresión con sangría

{"results":[{"gender":"female","name":
{"title":"Mrs","first":"Beatriz","last":"Páez"},"location":{"street":
{"number":743,"name":"Pasaje
Tamaulipas"},"city":"Kaua","state":"Campeche","country":"Mexico","postco
de":38941,"coordinates":
{"latitude":"43.6251","longitude":"-127.9305"},"timezone":
{"offset":"-3:00","description":"Brazil, Buenos Aires,
Georgetown"}},"email":"beatriz.paez@example.com","login":
{"uuid":"a651b615-64c2-451f-93a7-
68681ae74ce5","username":"redkoala487","password":"buttons","salt":"zsUj
ug7w","md5":"222db8c15ff2661514fb8b8e1534a2f8","sha1":"c10bf133d7609b4ff
5d8bec94cc3a5255210c918","sha256":"1842fd2d2f91c01823b4a166276c297ebd554
513c6406b4418b43dbff9051b23"},"dob":{"date":"1973-08-
29T18:24:48.618Z","age":52},"registered":{"date":"2010-03-
02T10:24:23.861Z","age":16},"phone":"(606) 943 7833","cell":"(613) 841
0077","id":{"name":"NSS","value":"63 18 55 9356 2"},"picture":
{"large":"https://randomuser.me/api/portraits/women/49.jpg","medium":"ht
tps://randomuser.me/api/portraits/med/women/49.jpg","thumbnail":"https:/
/randomuser.me/api/portraits/thumb/women/49.jpg"},"nat":"MX"}],"info":
{"seed":"3de1f576206e60fe","results":1,"page":1,"version":"1.4"}}



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## Parámetros Path

URLBase/endpoint/**{parametro}**

**CatAPI**

https://api.thecatapi.com/v1/images/0XYvRd7oD

https://api.thecatapi.com/v1/im... Chat

Impresión con sangría

{"id":"0XYvRd7oD","url":"https://cdn2.thecatapi.com/images/0XYvRd7oD.jpg
","breeds":[{"weight":{"imperial":"7 - 10","metric":"3 -
5"},"id":"abys","name":"Abyssinian","cfa_url":"http://cfa.org/Breeds/Bre
edsAB/Abyssinian.aspx","vetstreet_url":"http://www.vetstreet.com/cats/ab
yssinian","vcahospitals_url":"https://vcahospitals.com/know-your-
pet/cat-breeds/abyssinian","temperament":"Active, Energetic,
Independent, Intelligent,
Gentle","origin":"Egypt","country_codes":"EG","country_code":"EG","descr
iption":"The Abyssinian is easy to care for, and a joy to have in your
home. They're affectionate cats and love both people and other
animals.","life_span":"14 -
15","indoor":0,"lap":1,"alt_names":"","adaptability":5,"affection_level"
:5,"child_friendly":3,"dog_friendly":4,"energy_level":5,"grooming":1,"he
alth_issues":2,"intelligence":5,"shedding_level":2,"social_needs":5,"str
anger_friendly":5,"vocalisation":1,"experimental":0,"hairless":0,"natura
l":1,"rare":0,"rex":0,"suppressed_tail":0,"short_legs":0,"wikipedia_url"
:"https://en.wikipedia.org/wiki/Abyssinian_(cat)","hypoallergenic":0,"re
ference_image_id":"0XYvRd7oD"}],"width":1204,"height":1445}

PokeAPI

https://pokeapi.co/api/v2/pokemon/pikachu

https://pokeapi.co/api/v2/poke...

Impresión con sangría

{"abilities":[{"ability":
{"name":"static","url":"https://pokeapi.co/api/v2/ability/9/"},"is_hid
den":false,"slot":1},{"ability":{"name":"lightning-
rod","url":"https://pokeapi.co/api/v2/ability/31/"},"is_hidden":true,"
slot":3}],"base_experience":112,"cries":
{"latest":"https://raw.githubusercontent.com/PokeAPI/cries/main/cries/
pokemon/latest/25.ogg","legacy":"https://raw.githubusercontent.com/Pok
eAPI/cries/main/cries/pokemon/legacy/25.ogg"},"forms":
[{"name":"pikachu","url":"https://pokeapi.co/api/v2/pokemon-
form/25/"}],"game_indices":[{"game_index":84,"version":
{"name":"red","url":"https://pokeapi.co/api/v2/version/1/"}},
{"game_index":84,"version":
{"name":"blue","url":"https://pokeapi.co/api/v2/version/2/"}},
{"game_index":84,"version":
{"name":"yellow","url":"https://pokeapi.co/api/v2/version/3/"}},
{"game_index":25,"version":
{"name":"gold","url":"https://pokeapi.co/api/v2/version/4/"}},
{"game_index":25,"version":
{"name":"silver","url":"https://pokeapi.co/api/v2/version/5/"}},
{"game_index":25,"version":
{"name":"crystal","url":"https://pokeapi.co/api/v2/version/6/"}},
{"game_index":25,"version":
{"name":"ruby","url":"https://pokeapi.co/api/v2/version/7/"}},
{"game_index":25,"version":
{"name":"sapphire","url":"https://pokeapi.co/api/v2/version/8/"}},
{"game_index":25,"version":
{"name":"emerald","url":"https://pokeapi.co/api/v2/version/9/"}},
{"game_index":25,"version":
{"name":"firered","url":"https://pokeapi.co/api/v2/version/10/"}},
{"game_index":25,"version":



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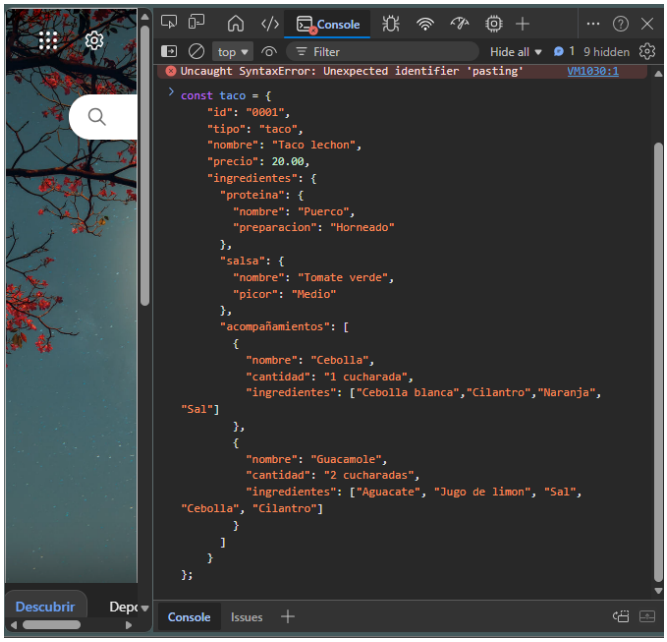
# Manipulación de JSON y el DOM

Código de objeto taco en la pestaña de consola del navegador.




```
> const taco = {
      "id": "0001",
      "tipo": "taco",
      "nombre": "Taco lechon",
      "precio": 20.00,
      "ingredientes": {
        "proteina": {
          "nombre": "Puerco",
          "preparacion": "Horneado"
        },
        "salsa": {
          "nombre": "Tomate verde",
          "picor": "Medio"
        },
        "acompañamientos": [
          {
            "nombre": "Cebolla",
            "cantidad": "1 cucharada",
            "ingredientes": ["Cebolla blanca","Cilantro","Naranja",
"Sal"]
          },
          {
            "nombre": "Guacamole",
            "cantidad": "2 cucharadas",
            "ingredientes": ["Aguacate", "Jugo de limon", "Sal",
"Cebolla", "Cilantro"]
          }
        ]
      }
};
```

## Ejercicios 1

1. Obtén el nombre de la proteína ("Puerco").

```
> console.log(taco.ingredientes.proteina.nombre);
Puerco                                                    VM233:1
```

2. Accede al primer elemento de la lista de ingredientes de la Cebolla ("Cebolla blanca").

```
> console.log(taco.ingredientes.acompañamientos[0].ingredientes[0]);
Cebolla blanca                                            VM362:1
```

3. Accede al último elemento de la lista de ingredientes del Guacamole ("Cilantro").

```
> console.log(taco.ingredientes.acompañamientos[1].ingredientes[4]);
Cilantro                                                  VM371:1
```



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**Ejercicio 2 Renderizar información dinámica desde un objeto JSON hacia el navegador utilizando JavaScript y manipulación del DOM.**

Código completo.

```html
<!--Por Luis Mario Ireta Xiu-->
<!doctype html>
<html lang="es">
  <head>
    <meta charset="UTF-8" />
    <title>Receta</title>
    <style>
      .card {
        border: 2px solid #2ecc71;
        border-radius: 10px;
        padding: 20px;
        width: 300px;
        font-family: sans-serif;
      }
      .precio {
        color: #e67e22;
        font-weight: bold;
        font-size: 1.2em;
      }
      ul {
        padding-left: 20px;
      }
    </style>
  </head>
  <body>
    <div id="taco-card" class="card">
      <h1 id="taco-nombre">Cargando...</h1>
      <p class="precio">Precio: <span id="taco-precio">0</span></p>
      <h3>Acompañamientos:</h3>
      <ul id="lista-acompañamientos"></ul>
    </div>

    <script>
      const taco = {
        id: "0001",
        tipo: "taco",
        nombre: "Taco lechon",
        precio: 20.0,
        ingredientes: {
          proteina: {
            nombre: "Puerco",
            preparacion: "Horneado",
          },
          salsa: {
            nombre: "Tomate verde",
            picor: "Medio",
          },
          acompañamientos: [
            {
              nombre: "Cebolla",
              cantidad: "1 cucharada",
```



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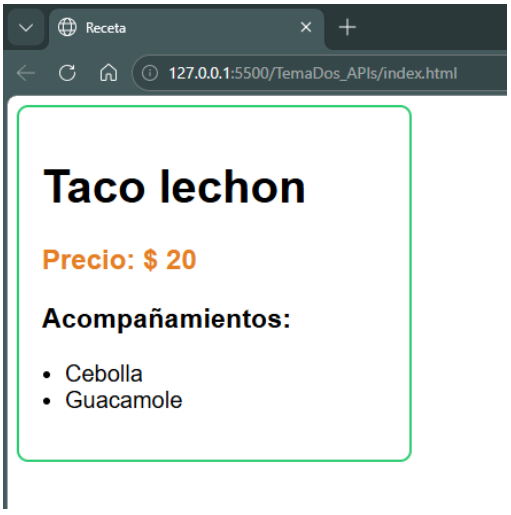
```
52                ingredientes: ["Cebolla blanca", "Cilantro", "Naranja", "Sal"]
53              },
54              {
55                nombre: "Guacamole",
56                cantidad: "2 cucharadas",
57                ingredientes: [
58                  "Aguacate",
59                  "Jugo de limon",
60                  "Sal",
61                  "Cebolla",
62                  "Cilantro",
63                ],
64              },
65            ],
66          },
67        };
68
69        // TU CÓDIGO AQUÍ:
70        let titulo = document.getElementById("taco-nombre");
71        let precio = document.getElementById("taco-precio");
72        let lista = document.getElementById("lista-acompañamientos");
73
74        // 1. Insertar nombre en el h1
75        titulo.textContent = taco.nombre;
76        // 2. Insertar precio con formato moneda
77        precio.textContent = `$ ${taco.precio}`;
78        // 3. Recorrer acompañamientos y crear <li> para cada uno
79        taco.ingredientes["acompañamientos"].forEach((taco) => {
80          let nuevo = document.createElement("li");
81          nuevo.textContent = taco.nombre;
82          lista.appendChild(nuevo);
83        });
84      </script>
85    </body>
86  </html>
```

Resultado.



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# JSON (JavaScript Object Notation)

## Ejercicio 1 Conversión de objeto de JS a JSON.

Código.

```javascript
//Por Luis Mario Ireta Xiu

let objetoJavaScript = {
  id: "0001",
  tipo: "taco",
  nombre: "Taco lechon",
  precio: 20.0,
  ingredientes: {
    proteina: {
      nombre: "Puerco",
      preparacion: "Horneado",
    },
    salsa: {
      nombre: "Tomate verde",
      picor: "Medio",
    },
    acompañamientos: [
      {
        nombre: "Cebolla",
        cantidad: "1 cucharada",
        ingredientes: ["Cebolla blanca", "Cilantro", "Naranja", "Sal"],
      },
      {
        nombre: "Guacamole",
        cantidad: "2 cucharadas",
        ingredientes: [
          "Aguacate",
          "Jugo de limon",
          "Sal",
          "Cebolla",
          "Cilantro",
        ],
      },
    ],
  },
};
//Serializar es convertir JSON
const jsonString = JSON.stringify(objetoJavaScript);

console.log(jsonString);
```

Resultado.

```
Luis@DESKTOP-0VPTEML MINGW64 ~/Proyecto/Programacion_Web_del_Lado_del_Servidor/TemaUno_Lenguajes/TemaDos_APIs/Ejercicio_2.1 (main)
$ node index.js
{"id":"0001","tipo":"taco","nombre":"Taco lechon","precio":20,"ingredientes":{"proteina":{"nombre":"Puerco","preparacion":"Horneado"},"salsa":{"nombre":"Tomate verde","picor":"Medio"},"acompañamientos":[{"nombre":"Cebolla","cantidad":"1 cucharada","ingredientes":["Cebolla blanca","Cilantro","Naranja","Sal"]},{"nombre":"Guacamole","cantidad":"2 cucharadas","ingredientes":["Aguacate","Jugo de limon","Sal","Cebolla","Cilantro"]}]}}
```



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## Ejercicio 2 Conversión de JSON a objeto de JS.

Código.

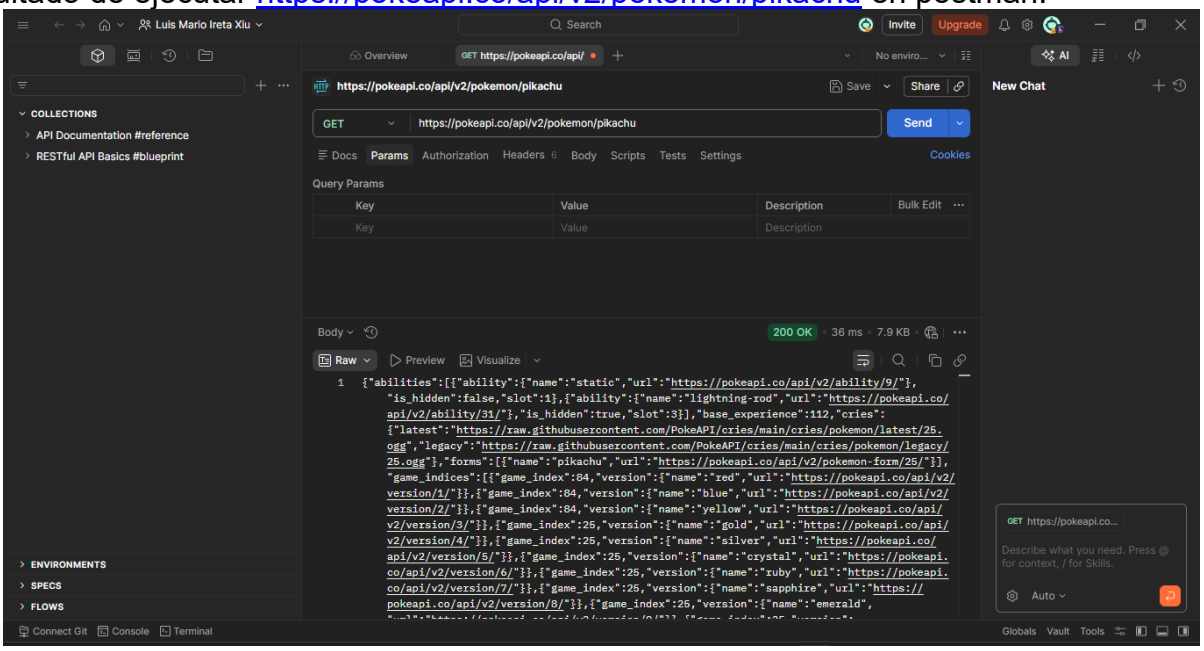
```
//Por Luis Mario Ireta Xiu

// Cadena JSON
const jsonString = '{"nombre":"Taco de Pollo","ingredientes": {"proteina":"Pollo","salsa":"Salsa Verde"}}';

// Deserializar para convertir a objeto JavaScript
const objetoDeserealizado = JSON.parse(jsonString);

console.log(objetoDeserealizado);
```

Resultado.

```
Luis@DESKTOP-0VPTEML MINGW64 ~/Proyecto/Programacion_Web_del_Lado_del_Servidor/TemaUno_Lenguajes/TemaDos_APIs/Ejercicio_2.1 (main)
$ node index2.js
{
  nombre: 'Taco de Pollo',
  ingredientes: { proteina: 'Pollo', salsa: 'Salsa Verde' }
}
```

Visualización de JSON

Resultado de ejecutar https://pokeapi.co/api/v2/pokemon/pikachu en postman.



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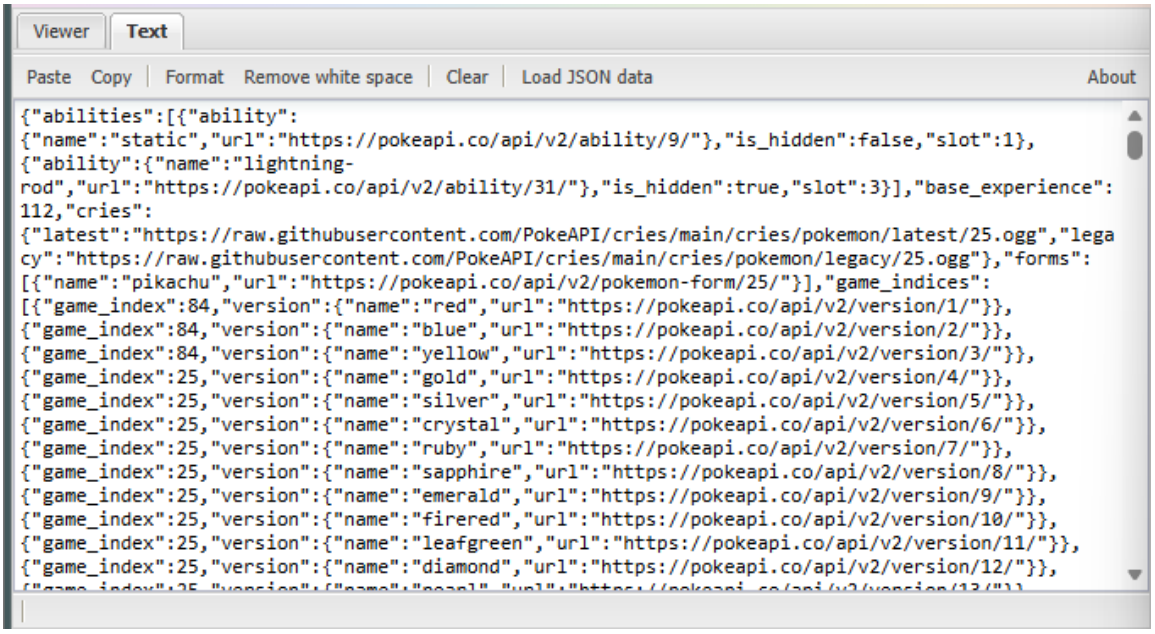
Después de abrir la pestaña Viewer e interpretar la información obtenida.

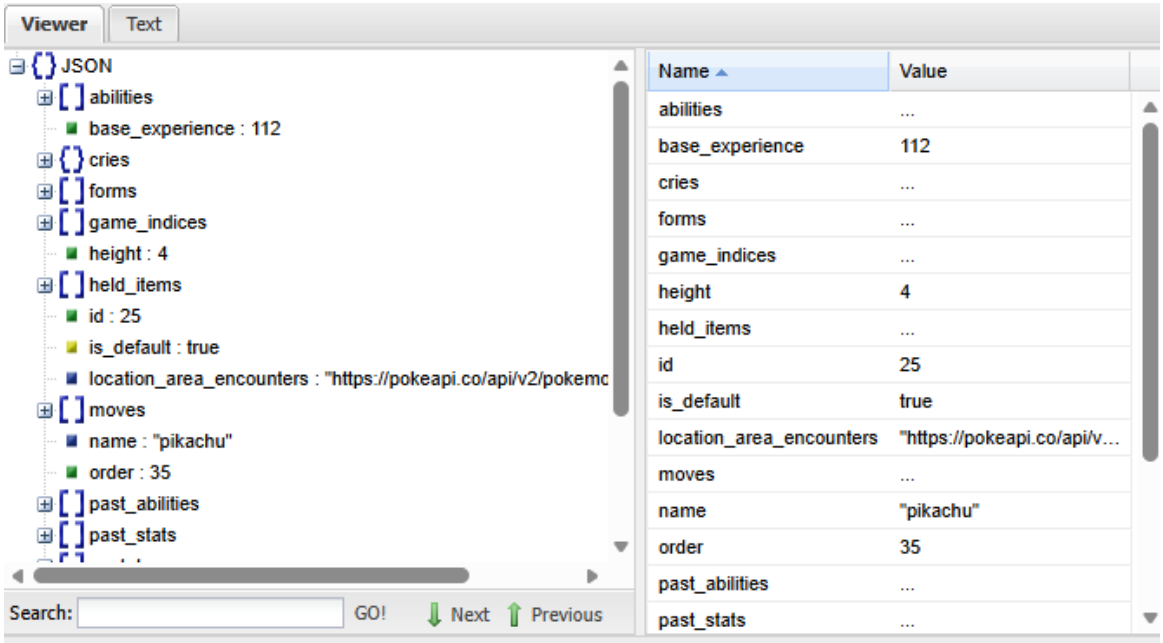
Antes.





```
{"abilities":[{"ability":
{"name":"static","url":"https://pokeapi.co/api/v2/ability/9/"},"is_hidden":false,"slot":1},
{"ability":{"name":"lightning-
rod","url":"https://pokeapi.co/api/v2/ability/31/"},"is_hidden":true,"slot":3}],"base_experience":
112,"cries":
{"latest":"https://raw.githubusercontent.com/PokeAPI/cries/main/cries/pokemon/latest/25.ogg","lega
cy":"https://raw.githubusercontent.com/PokeAPI/cries/main/cries/pokemon/legacy/25.ogg"},"forms":
[{"name":"pikachu","url":"https://pokeapi.co/api/v2/pokemon-form/25/"}],"game_indices":
[{"game_index":84,"version":{"name":"red","url":"https://pokeapi.co/api/v2/version/1/"}},
{"game_index":84,"version":{"name":"blue","url":"https://pokeapi.co/api/v2/version/2/"}},
{"game_index":84,"version":{"name":"yellow","url":"https://pokeapi.co/api/v2/version/3/"}},
{"game_index":25,"version":{"name":"gold","url":"https://pokeapi.co/api/v2/version/4/"}},
{"game_index":25,"version":{"name":"silver","url":"https://pokeapi.co/api/v2/version/5/"}},
{"game_index":25,"version":{"name":"crystal","url":"https://pokeapi.co/api/v2/version/6/"}},
{"game_index":25,"version":{"name":"ruby","url":"https://pokeapi.co/api/v2/version/7/"}},
{"game_index":25,"version":{"name":"sapphire","url":"https://pokeapi.co/api/v2/version/8/"}},
{"game_index":25,"version":{"name":"emerald","url":"https://pokeapi.co/api/v2/version/9/"}},
{"game_index":25,"version":{"name":"firered","url":"https://pokeapi.co/api/v2/version/10/"}},
{"game_index":25,"version":{"name":"leafgreen","url":"https://pokeapi.co/api/v2/version/11/"}},
{"game_index":25,"version":{"name":"diamond","url":"https://pokeapi.co/api/v2/version/12/"}},
```


Después.



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Información del archivo recetaTacos.json.

```json
[
  {
    "id": "0002",
    "tipo": "taco",
    "nombre": "Taco de cochinita pibil",
    "precio": 25.00,
    "ingredientes": {
      "proteina": {
        "nombre": "Cerdo",
        "preparacion": "Marinado con achiote y cocido en pib"
      },
      "salsa": {
        "nombre": "Xnipec",
        "picor": "Alto"
      },
      "acompañamientos": [
        {
          "nombre": "Cebolla morada encurtida",
          "cantidad": "1 porción",
          "ingredientes": ["Cebolla morada", "Naranja agria", "Orégano", "Sal"]
        },
        {
          "nombre": "Tortilla de maíz",
          "cantidad": "2 piezas",
          "ingredientes": ["Masa de maíz", "Agua", "Sal"]
        }
      ]
    }
  },
```

```json
  {
    "id": "0003",
    "tipo": "taco",
    "nombre": "Taco de lechón al horno",
    "precio": 30.00,
    "ingredientes": {
      "proteina": {
        "nombre": "Lechón",
        "preparacion": "Horneado lentamente"
      },
      "salsa": {
        "nombre": "Salsa de chile habanero",
        "picor": "Muy alto"
      },
      "acompañamientos": [
        {
          "nombre": "Repollo",
          "cantidad": "1 porción",
          "ingredientes": ["Repollo", "Sal", "Limón"]
        },
        {
          "nombre": "Frijol colado",
          "cantidad": "1 cucharada",
          "ingredientes": ["Frijol negro", "Manteca", "Sal"]
        }
      ]
    }
  },
```



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```
  {
    "id": "0004",
    "tipo": "taco",
    "nombre": "Taco de relleno negro",
    "precio": 28.00,
    "ingredientes": {
      "proteina": {
        "nombre": "Pavo",
        "preparacion": "Cocido en recado negro"
      },
      "salsa": {
        "nombre": "Salsa de tomate",
        "picor": "Medio"
      },
      "acompañamientos": [
        {
          "nombre": "Huevo cocido",
          "cantidad": "1 pieza",
          "ingredientes": ["Huevo", "Sal"]
        },
        {
          "nombre": "Tortilla de maíz",
          "cantidad": "2 piezas",
          "ingredientes": ["Masa de maíz", "Agua", "Sal"]
        }
      ]
    }
  },
```

```
  },
  {
    "id": "0005",
    "tipo": "taco",
    "nombre": "Taco de pollo pibil",
    "precio": 24.00,
    "ingredientes": {
      "proteina": {
        "nombre": "Pollo",
        "preparacion": "Marinado con achiote y horneado"
      },
      "salsa": {
        "nombre": "Salsa de tomate con habanero",
        "picor": "Medio"
      },
      "acompañamientos": [
        {
          "nombre": "Cebolla encurtida",
          "cantidad": "1 porción",
          "ingredientes": ["Cebolla morada", "Vinagre", "Sal"]
        },
        {
          "nombre": "Arroz",
          "cantidad": "1 porción",
          "ingredientes": ["Arroz", "Ajo", "Sal"]
        }
      ]
    }
  }
]
```



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# Aplicación usando datos JSON y API Fetch.

Creación de proyecto.

```
TemaDos_APIs\Ejercicio_2.1
  AppRecetaTacos
```

Proyecto con archivo index.js.

```
JS index.js                U
```

Se añade el archivo recetaTacos.json.

```
{} recetaTacos.json        U
```

Inicializar npm.

```
Luis@DESKTOP-0VPTEML MINGW64 ~/Proyecto/Programacion_Web
_del_Lado_del_Servidor/TemaUno_Lenguajes/TemaDos_APIs/Ej
ercicio_2.1/AppRecetaTacos (main)
$ npm init
This utility will walk you through creating a package.js
on file.
It only covers the most common items, and tries to guess
 sensible defaults.

See `npm help init` for definitive documentation on thes
e fields
and exactly what they do.

Use `npm install <pkg>` afterwards to install a package
and
save it as a dependency in the package.json file.

Press ^C at any time to quit.
package name: (apprecetatacos)
version: (1.0.0)
description:
entry point: (index.js)
test command:
git repository:
keywords:
author: LIX1313
license: (ISC)
type: (commonjs)
About to write to C:\Users\Luis\Proyecto\Programacion_We
b_del_Lado_del_Servidor\TemaUno_Lenguajes\TemaDos_APIs\E
jercicio_2.1\AppRecetaTacos\package.json:

{
  "name": "apprecetatacos",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1
"
```



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Agregar module.

```
emaDos_APIs > Ejercicio_2.1 > AppRecetaTacos > {} package.json > abc type
{
  "name": "apprecetatacos",
  "version": "1.0.0",
  "description": "",
  "license": "ISC",
  "author": "LIX1313",
  "type": "module",
  "main": "index.js",
```

Instalar framework express y el middleware body-parser.

```
Luis@DESKTOP-0VPTEML MINGW64 ~/P
royecto/Programacion_Web_del_Lad
o_del_Servidor/TemaUno_Lenguajes
/TemaDos_APIs/Ejercicio_2.1/AppR
ecetaTacos (main) …
$ npm install express

added 65 packages, and audited 6
6 packages in 4s

22 packages are looking for fund
ing
  run `npm fund` for details

found 0 vulnerabilities

Luis@DESKTOP-0VPTEML MINGW64 ~/P
royecto/Programacion_Web_del_Lad
o_del_Servidor/TemaUno_Lenguajes
/TemaDos_APIs/Ejercicio_2.1/AppR
ecetaTacos (main)
$ nom install body-parser
bash: nom: command not found

Luis@DESKTOP-0VPTEML MINGW64 ~/P
royecto/Programacion_Web_del_Lad
o_del_Servidor/TemaUno_Lenguajes
/TemaDos_APIs/Ejercicio_2.1/AppR
ecetaTacos (main)
$ npm install body-parser

up to date, audited 66 packages
in 2s

22 packages are looking for fund
ing
  run `npm fund` for details

found 0 vulnerabilities
```



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Boilerplate de la aplicación.

```
import express from "express";
import bodyParser from "body-parser";

const app = express();
const port = 3000;


app.use(bodyParser.urlencoded({ extended: true })); //Habilitar el middleware

//Obtener datos
app.get("/", (req, res) => {
  app.send("Hola");
});

app.listen(port, () => {
  console.log("El servidor se ejecuta en: " + port);
});

```

Carpeta public.

## Boilerplate para un documento html terminado.

```html
<!doctype html>
<html>
  <head>
    <meta charset="utf-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <title>TacosMx</title>
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <link rel="stylesheet" type="text/css" media="screen" href="estilo.css" />
  </head>
  <body>
    <h1>Tacoschidos 🫓</h1>
    <form id="form" class="buttons">
      <button type="button" data-type="Cerdo">🐷</button>
      <button type="button" data-type="Lechón">🐖</button>
      <button type="button" data-type="Pavo">🦃</button>
      <button type="button" data-type="Pollo">🐤</button>
    </form>
    <div id="Container"></div>
    <script>
      document.querySelectorAll("#form button").forEach((button) => {
        button.addEventListener("click", async (event) => {
          const type = event.target.getAttribute("data-type").toLowerCase();

          const response = await fetch(`/receta/${type}`);
          const receta = await response.json();

          const contenedor = document.getElementById("Container");
          if (receta.error) {
            contenedor.innerHTML = "<h2>Receta no encontrada</h2>"; //si no se recupera el contenido HTML de un elemento.
          } else {
            contenedor.innerHTML = `
            <h2>${receta.nombre}</h2>
            <h3>Ingredientes:</h3>
```

```html
          <ul id="ingredientesLista">
            <li>Carne de: ${receta.ingredientes.proteina.nombre}, Modo preparación: ${receta.ingredientes.proteina.preparacion}</li>
            <li> Salsa: ${receta.ingredientes.salsa.nombre} (${receta.ingredientes.salsa.picor})</li>
            ${receta.ingredientes.acompañamientos.map((acompañamiento) => `<li> Acompañamiento: ${acompañamiento.cantidad} : ${acompañamiento.nombre}</li>`).join("")}</ul>`;
          }
        });
      });
    </script>
  </body>
</html>
```



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## Código JS de Backend terminado.

```javascript
import express from "express";
import bodyParser from "body-parser";

const app = express();
const port = 3000;

const recetaJSON = `[
  {
    "id": "0002",
    "tipo": "taco",
    "nombre": "Taco de cochinita pibil",
    "precio": 25.00,
    "ingredientes": {
      "proteina": {
        "nombre": "Cerdo",
        "preparacion": "Marinado con achiote y cocido en pib"
      },
      "salsa": {
        "nombre": "Xnipec",
        "picor": "Alto"
      },
      "acompañamientos": [
        {
          "nombre": "Cebolla morada encurtida",
          "cantidad": "1 porción",
          "ingredientes": ["Cebolla morada", "Naranja agria", "Orégano", "Sal"]
        },
        {
          "nombre": "Tortilla de maíz",
          "cantidad": "2 piezas",
          "ingredientes": ["Masa de maíz", "Agua", "Sal"]
        }
      ]
    }
  },
  {
    "id": "0003",
    "tipo": "taco",
    "nombre": "Taco de lechón al horno",
    "precio": 30.00,
    "ingredientes": {
      "proteina": {
        "nombre": "Lechón",
        "preparacion": "Horneado lentamente"
      },
      "salsa": {
        "nombre": "Salsa de chile habanero",
        "picor": "Muy alto"
      },
      "acompañamientos": [
        {
          "nombre": "Repollo",
          "cantidad": "1 porción",
          "ingredientes": ["Repollo", "Sal", "Limón"]
        },
        {
          "nombre": "Frijol colado",
          "cantidad": "1 cucharada",
          "ingredientes": ["Frijol negro", "Manteca", "Sal"]
        }
      ]
    }
  },
```



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```
64    {
65      "id": "0004",
66      "tipo": "taco",
67      "nombre": "Taco de relleno negro",
68      "precio": 28.00,
69      "ingredientes": {
70        "proteina": {
71          "nombre": "Pavo",
72          "preparacion": "Cocido en recado negro"
73        },
74        "salsa": {
75          "nombre": "Salsa de tomate",
76          "picor": "Medio"
77        },
78        "acompañamientos": [
79          {
80            "nombre": "Huevo cocido",
81            "cantidad": "1 pieza",
82            "ingredientes": ["Huevo", "Sal"]
83          },
84          {
85            "nombre": "Tortilla de maíz",
86            "cantidad": "2 piezas",
87            "ingredientes": ["Masa de maíz", "Agua", "Sal"]
88          }
89        ]
90      }
91    },
92    {
93      "id": "0005",
94      "tipo": "taco",
95      "nombre": "Taco de pollo pibil",
96      "precio": 24.00,
```

```
7   const recetaJSON = `[
97      "ingredientes": {
98        "proteina": {
99          "nombre": "Pollo",
100         "preparacion": "Marinado con achiote y horneado"
101       },
102       "salsa": {
103         "nombre": "Salsa de tomate con habanero",
104         "picor": "Medio"
105       },
106       "acompañamientos": [
107         {
108           "nombre": "Cebolla encurtida",
109           "cantidad": "1 porción",
110           "ingredientes": ["Cebolla morada", "Vinagre", "Sal"]
111         },
112         {
113           "nombre": "Arroz",
114           "cantidad": "1 porción",
115           "ingredientes": ["Arroz", "Ajo", "Sal"]
116         }
117       ]
118     }
119   }
120 ]`;
121
122 const recetasTacos = JSON.parse(recetaJSON);
123
124 app.use(bodyParser.urlencoded({ extended: true })); //Habilitar el middleware
125 app.use(express.static("public")); //Acceso a estilos
126 app.use(bodyParser.json());
```

```
128   //Obtener datos
129   app.get("/receta/:type", (req, res) => {
130     const elegirTaco = recetasTacos.find(
131       (r) =>
132         r.ingredientes.proteina.nombre.toLowerCase() ===
133         req.params.type.toLowerCase(),
134     );
135     res.json(elegirTaco || { error: "Receta no encontrada" });
136   });
137
138   app.listen(port, () => {
139     console.log("El servidor se ejecuta en: " + port);
140   });
```



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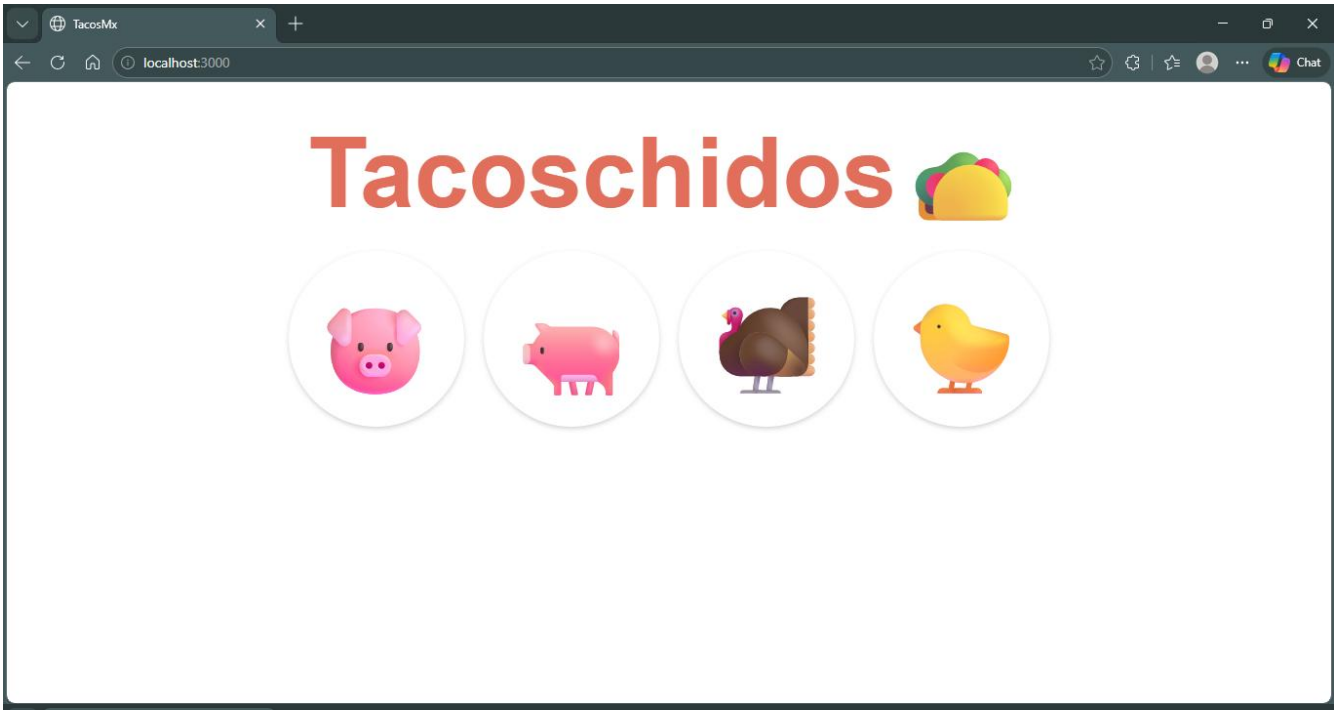
**Resultado.**



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